

Norway

SLEIPNER

Volve F

F-1

F-1 A

Design: F-1 A

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 A	Survey Calculation Method:	Minimum Curvature
Design:	F-1 A	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-1					
Well Position	+N/-S	3.17 m	Northing:	6,478,566.69 m	Latitude:	58° 26' 29.9072 N
	+E/-W	-3.53 m	Easting:	435,046.49 m	Longitude:	1° 53' 14.7075 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-1 A				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	18.08.2013	-1.14	71.62	50,520

Design	F-1 A				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	2,587.00
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	3.17	-3.53	337.69	

Survey Program	Date	29.01.2015			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
150.15	210.70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)	
237.60	358.50	15/9 F1 26" Gyro Scientific Modular (F-1)	NOT_useMWD gyro, Baker Hughes' MWD gyro service (GyroTrak)		
378.00	737.40	15/9 F-1 26" SDI Keeper Gyro Single Shot	Gyro, old	Other Gyro Tools	
756.80	1,335.60	15/9 F-1 26" OnTrak (F-1)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
1,387.10	2,587.00	15/9 F-1 17 1/2" AutoTrak G3 (F-1)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
2,621.10	3,671.10	15/9 F-1 A 8 1/2" Pilot AutoTrak (F-1 A)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	3.17	-3.53	0.00	0.000	0.000	0.000	
150.15	0.00	0.00	150.15	3.17	-3.53	0.00	0.000	0.000	0.000	
160.45	0.00	0.00	160.45	3.17	-3.53	0.00	0.000	0.000	0.000	
170.04	0.00	0.00	170.04	3.17	-3.53	0.00	0.000	0.000	0.000	
179.79	0.00	0.00	179.79	3.17	-3.53	0.00	0.000	0.000	0.000	
189.74	0.00	0.00	189.74	3.17	-3.53	0.00	0.000	0.000	0.000	

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Design:	F-1 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
198.50	0.00	0.00	198.50	3.17	-3.53	0.00	0.000	0.000	0.000	
210.70	0.00	0.00	210.70	3.17	-3.53	0.00	0.000	0.000	0.000	
237.60	1.84	302.24	237.60	3.40	-3.90	0.35	2.052	2.052	0.000	
251.10	1.91	304.70	251.09	3.64	-4.27	0.72	0.237	0.156	5.467	
264.90	2.47	305.76	264.88	3.94	-4.70	1.16	1.221	1.217	2.304	
279.20	2.05	306.81	279.17	4.28	-5.15	1.64	0.885	-0.881	2.203	
292.70	1.80	304.00	292.66	4.54	-5.52	2.03	0.594	-0.556	-6.244	
302.80	1.13	330.72	302.76	4.72	-5.70	2.26	2.791	-1.990	79.366	
315.50	1.41	333.18	315.45	4.96	-5.83	2.54	0.674	0.661	5.811	
332.20	2.40	314.19	332.14	5.39	-6.18	3.06	2.086	1.778	-34.114	
342.20	2.78	301.18	342.13	5.66	-6.54	3.45	2.093	1.140	-39.030	
358.50	3.30	293.10	358.41	6.05	-7.30	4.10	1.238	0.957	-14.871	
378.00	3.07	298.60	377.88	6.52	-8.28	4.91	0.588	-0.354	8.462	
414.80	3.30	293.64	414.62	7.42	-10.11	6.43	0.292	0.187	-4.043	
455.40	3.07	292.72	455.16	8.31	-12.19	8.04	0.174	-0.170	-0.680	
495.40	2.84	290.41	495.11	9.07	-14.10	9.47	0.194	-0.172	-1.732	
535.60	2.77	293.07	535.26	9.79	-15.93	10.84	0.110	-0.052	1.985	
575.80	3.06	287.04	575.41	10.49	-17.85	12.21	0.315	0.216	-4.500	
616.50	4.01	305.32	616.03	11.63	-20.05	14.10	1.078	0.700	13.474	
656.80	6.13	317.80	656.17	14.04	-22.65	17.32	1.770	1.578	9.290	
697.10	8.24	324.71	696.15	17.99	-25.76	22.15	1.695	1.571	5.144	
720.00	9.57	322.37	718.78	20.84	-27.87	25.59	1.805	1.742	-3.066	
737.40	10.71	320.81	735.91	23.24	-29.78	28.53	2.022	1.966	-2.690	
756.80	10.39	320.18	754.98	25.98	-32.04	31.92	0.526	-0.495	-0.974	
798.90	9.88	320.50	796.42	31.68	-36.76	39.00	0.366	-0.363	0.228	
838.30	10.04	315.72	835.23	36.75	-41.31	45.41	0.641	0.122	-3.640	
868.00	11.16	309.73	864.42	40.44	-45.33	50.35	1.586	1.131	-6.051	
878.20	11.28	310.41	874.43	41.72	-46.85	52.11	0.525	0.353	2.000	
918.00	10.70	315.76	913.50	46.89	-52.39	59.00	0.884	-0.437	4.033	
999.00	8.88	319.88	993.32	57.06	-61.67	71.92	0.722	-0.674	1.526	
1,079.80	9.07	320.34	1,073.13	66.73	-69.75	83.94	0.075	0.071	0.171	
1,120.40	11.38	312.16	1,113.08	71.88	-74.76	90.61	2.012	1.707	-6.044	
1,160.80	11.86	307.61	1,152.65	77.09	-81.01	97.80	0.768	0.356	-3.379	
1,201.00	11.69	307.74	1,192.01	82.10	-87.50	104.90	0.128	-0.127	0.097	
1,241.40	12.44	309.36	1,231.51	87.37	-94.10	112.28	0.611	0.557	1.203	
1,281.80	12.06	309.62	1,270.99	92.82	-100.72	119.83	0.285	-0.282	0.193	
1,322.40	10.85	310.25	1,310.79	97.99	-106.90	126.97	0.899	-0.894	0.466	
1,335.60	10.17	309.34	1,323.76	99.53	-108.75	129.10	1.591	-1.545	-2.068	
1,387.10	7.66	329.93	1,374.65	105.39	-113.99	136.50	2.343	-1.462	11.994	
1,428.00	10.47	345.74	1,415.04	111.35	-116.27	142.88	2.736	2.061	11.597	
1,468.30	12.18	6.35	1,454.57	119.13	-116.70	150.24	3.249	1.273	15.342	
1,508.70	13.62	22.56	1,493.96	127.76	-114.40	157.36	2.881	1.069	12.037	
1,551.30	15.49	29.92	1,535.20	137.33	-109.64	164.40	1.850	1.317	5.183	
1,589.10	17.80	32.61	1,571.41	146.57	-104.01	170.81	1.932	1.833	2.135	

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Wellbore:	F-1 A	Survey Calculation Method:	Minimum Curvature
Design:	F-1 A	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,629.50	20.24	33.68	1,609.60	157.59	-96.80	178.27	1.830	1.812	0.795
1,663.40	23.45	33.72	1,641.07	168.08	-89.81	185.32	2.841	2.841	0.035
1,710.40	26.64	33.77	1,683.64	184.62	-78.76	196.43	2.036	2.036	0.032
1,750.30	29.38	36.44	1,718.87	199.94	-67.97	206.50	2.265	2.060	2.008
1,790.40	29.98	36.55	1,753.70	215.90	-56.16	216.79	0.451	0.449	0.082
1,831.10	30.06	36.61	1,788.94	232.25	-44.02	227.31	0.063	0.059	0.044
1,871.10	30.03	36.77	1,823.57	248.31	-32.06	237.62	0.064	-0.022	0.120
1,911.60	29.86	35.39	1,858.66	264.65	-20.15	248.22	0.526	-0.126	-1.022
1,952.20	29.78	34.81	1,893.89	281.16	-8.54	259.09	0.221	-0.059	-0.429
1,992.80	29.92	35.24	1,929.10	297.71	3.06	270.00	0.189	0.103	0.318
2,027.30	29.89	35.13	1,959.01	311.77	12.97	279.24	0.054	-0.026	-0.096
2,066.30	29.92	34.80	1,992.82	327.70	24.11	289.75	0.129	0.023	-0.254
2,113.70	29.80	35.71	2,033.92	346.98	37.73	302.41	0.297	-0.076	0.576
2,154.70	29.85	34.79	2,069.49	363.63	49.50	313.35	0.337	0.037	-0.673
2,194.30	29.84	36.98	2,103.84	379.59	61.05	323.73	0.826	-0.008	1.659
2,234.70	29.79	38.27	2,138.89	395.50	73.31	333.80	0.478	-0.037	0.958
2,275.10	30.01	37.36	2,173.92	411.41	85.66	343.83	0.374	0.163	-0.676
2,305.90	29.96	35.61	2,200.60	423.78	94.81	351.80	0.853	-0.049	-1.705
2,396.50	30.06	35.35	2,279.05	460.68	121.11	375.96	0.054	0.033	-0.086
2,436.70	30.19	35.59	2,313.82	477.12	132.82	386.72	0.132	0.097	0.179
2,470.80	30.22	36.84	2,343.29	490.96	142.95	395.67	0.554	0.026	1.100
2,517.20	30.25	36.59	2,383.38	509.69	156.92	407.70	0.084	0.019	-0.162
2,557.90	30.19	34.71	2,418.55	526.33	168.86	418.56	0.699	-0.044	-1.386
2,587.00	30.19	33.06	2,443.70	538.48	177.02	426.71	0.855	0.000	-1.701
2,621.10	29.93	25.32	2,473.23	553.36	185.34	437.31	3.417	-0.229	-6.809
2,647.40	28.62	18.88	2,496.17	565.25	190.18	446.48	3.889	-1.494	-7.346
2,688.10	27.17	11.08	2,532.16	583.60	195.12	461.58	2.892	-1.069	-5.749
2,727.50	26.91	3.24	2,567.26	601.34	197.36	477.14	2.719	-0.198	-5.970
2,768.60	26.81	353.59	2,603.94	619.84	196.85	494.45	3.180	-0.073	-7.044
2,809.10	28.15	343.98	2,639.89	638.11	193.19	512.74	3.428	0.993	-7.119
2,849.40	30.20	333.38	2,675.10	656.32	186.02	532.32	4.132	1.526	-7.891
2,889.60	32.18	322.06	2,709.52	673.82	174.90	552.73	4.611	1.478	-8.448
2,930.40	34.72	315.89	2,743.57	690.74	160.12	573.99	3.119	1.868	-4.537
2,970.70	38.45	310.87	2,775.93	707.19	142.65	595.84	3.558	2.777	-3.737
3,010.90	42.29	305.91	2,806.56	723.31	122.23	618.50	3.735	2.866	-3.701
3,051.00	45.11	300.16	2,835.56	738.37	99.01	641.25	3.643	2.110	-4.302
3,091.40	48.20	296.12	2,863.29	752.20	73.10	663.88	3.165	2.295	-3.000
3,130.50	53.44	294.14	2,887.99	765.04	45.66	686.18	4.189	4.020	-1.519
3,171.20	57.55	289.29	2,911.05	777.41	14.51	709.44	4.224	3.029	-3.575
3,212.10	62.02	285.32	2,931.63	787.89	-19.22	731.95	4.132	3.279	-2.912
3,251.60	63.21	281.36	2,949.80	795.98	-53.34	752.38	2.819	0.904	-3.008
3,292.00	58.32	281.29	2,969.53	802.90	-87.89	771.90	3.631	-3.631	-0.052
3,333.10	53.03	282.89	2,992.70	809.99	-121.07	791.05	3.980	-3.861	1.168

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Design:	F-1 A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,373.40	46.91	287.36	3,018.61	817.98	-150.85	809.75	5.218	-4.556	3.328	
3,413.80	44.30	285.04	3,046.88	826.05	-178.56	827.73	2.296	-1.938	-1.723	
3,454.10	43.58	283.54	3,075.90	832.95	-205.66	844.40	0.942	-0.536	-1.117	
3,494.30	44.11	283.07	3,104.89	839.36	-232.76	860.62	0.464	0.396	-0.351	
3,534.60	44.10	283.10	3,133.83	845.71	-260.08	876.86	0.017	-0.007	0.022	
3,574.30	44.13	283.44	3,162.33	852.05	-286.97	892.94	0.180	0.023	0.257	
3,615.20	44.05	284.07	3,191.70	858.82	-314.61	909.69	0.327	-0.059	0.462	
3,655.40	44.08	285.06	3,220.59	865.85	-341.67	926.47	0.514	0.022	0.739	
3,671.10	44.05	285.44	3,231.87	868.72	-352.21	933.13	0.508	-0.057	0.726	
3,682.00	44.05	285.44	3,239.71	870.74	-359.51	937.77	0.000	0.000	0.000	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.90	220.90	30"	30.000	36.000	
1,348.40	1,336.38	20"	20.000	26.000	
2,595.40	2,450.97	13 3/8"	13.375	17.500	

Design Annotations					
Measured Depth (m)	Vertical Depth (m)	Local Coordinates		Comment	
		+N/-S (m)	+E/-W (m)		
3,682.00	3,239.71	870.74	-359.51	Projection to TD.	

Checked By: _____ Approved By: _____ Date: _____